What is unified threat management (UTM)?

Unified threat management (UTM) describes an information security (infosec) system that provides a single point of protection against threats, including viruses, worms, spyware and other malware, and network attacks. It combines security, performance, management and compliance capabilities into a single installation, making it easier for administrators to manage networks.

Unlike antivirus tools, a UTM system does not just protect personal computers (PCs) and servers. It protects an entire network and individual users by scanning all network traffic, filtering potentially dangerous content and blocking intrusions. Many small and medium-sized businesses (SMBs) have adopted UTM systems, finding it easier to handle their infosec with a single system, rather than several smaller ones.

UTM systems combine multiple security features into a single device or software program. This can help because there are five primary kinds of threats that organizations need to protect against:

1. *malware*
2. *phishing and social engineering*
3. *viruses, worms and Trojans*
4. *hackers*

5. *denial of service (DoS)*

When dealing with these threats, a separate technology is typically required to resolve each issue. That ends up being more complicated than it needs to be, which is why UTM systems exist.

UTM and next-generation firewalls (NGFWs) are both firewall technologies serving similar purposes, but they're also different in some key areas. NGFWs were originally developed to fill network security gaps left by traditional firewalls and include application intelligence and an intrusion prevention system (IPS), as well as DoS protection. UTM refers to the ability of a single device to perform the functions of an NGFW, firewall and virtual private network (VPN), while an NGFW is a network security platform that provides a gateway between internal and external networks. The major difference between these two firewall types is that a UTM system typically offers more features than an NGFW, such as an intrusion detection system (IDS) and spam filtering, since it is able to monitor and protect internal networks from intruders.

Learn more about the differences between unified threat management and next-generation firewalls here.



How UTM works

Understanding threats and identifying weaknesses to an organization's network are critical for security. A UTM system can help accomplish this by using two inspection methods that address different types of threats:

1. ***Flow-based inspection.*** *Flow-based inspection, also known as stream-based inspection, samples data that enters a network security device, such as a firewall or IPS. The devices inspect the data for malicious activity, such as viruses, intrusions and other hacking attempts.*
2. ***Proxy-based inspection.*** *Proxy-based inspection is a network security technique that can be used to examine the contents of packets that pass into and out of a network security device, such as a firewall, IPS or VPN server. By using a proxy server to inspect these packets, the network security device can act as a proxy to reconstruct the content entering the device.*

Unified threat management devices

UTM devices are hardware or software that tie together network security features into one simple-to-use, easy-to-manage appliance. In addition to having a firewall, VPN and IPS, every UTM appliance supports network- or cloud-based centralized management. For example, Cisco Meraki appliances use a cloud-based management tool that can be deployed remotely on a per-device basis.

Unified threat management features

UTMs typically include the following security features.

Antispam services

Antispam services or spam filters are designed to block or tag incoming email-based attacks by scanning inbound and outbound email traffic for signs of a possible attack. Antispam systems use algorithms to detect spam by scanning message content for patterns that are associated with spam. Some systems look for certain words, others for specific language patterns and others for whole word patterns using a process called Bayesian analysis. If the message appears to be spam or malware, the contents are tagged or quarantined.

URL filtering and application control

UTM devices can perform many functions that help secure a corporation or other organization's network, including Uniform Resource Locator (URL) filtering and application control. With application control, a UTM device can put specific applications on an allowlist so they can connect to the internet without dealing with spam content filtering or other security measures. Application control is usually combined with a UTM device's firewall and other features to ensure that all traffic entering the corporate network is protected.

Firewalls

A firewall is a hardware- or software-based security measure that restricts access to a private network by monitoring incoming and outgoing traffic between different networks. It keeps unauthorized -- or malicious -- users from gaining access to data or resources such as file servers, printers and web servers. There are three main types of firewalls: packet filtering, circuit-level gateway and application-level gateway.

Intrusion detection systems and intrusion prevention systems

An IDS monitors the network for signs of a cyber attack, while an IPS takes action to stop attacks by neutralizing malicious traffic.

The goal of an IDS is to detect abnormal behavior so that it can be analyzed, recorded and reported. It can't actually block any incoming threats, but it can notify an administrator about an intrusion and log the activity for later analysis. An IPS, on the other hand, is a type of security technology that can alter

network traffic to block malicious activities. An IPS feature can be added to an existing IDS or firewall.

VPN

The role of a VPN is to create a secure connection between two computers over a public network. This enables file sharing securely between co-workers, accessing data remotely or using any number of other services without fear that an outside party will intercept the data. VPNs work by using encryption to protect data from unauthorized access when crossing between public and private networks, thereby creating a secure connection that is encrypted within a tunnel over the public internet.

Content filtering

Web content filtering is a method of controlling what types of information can pass into or out of a network, using various filtering methods, such as by Internet Protocol (IP) address, port number or media access control (MAC) address. Content filtering is used on networks to block unwanted content and to protect against data loss by filtering outgoing data to prevent sensitive information from being transmitted.

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*Best unified***Benefits of Using a Unified Threat Management Solution**

Flexibility and adaptability

With a UTM network, you can use a set of flexible solutions to handle the complicated assortment of networking setups available in modern business infrastructure. You can cherry-pick what you need from a selection of security management tools, choosing what is best for your specific network. You can also opt to obtain one licensing model that comes with all the technologies you want, saving you time shopping for individual solutions.

Because a UTM is flexible, you have the freedom to deploy more than one security technology as you see fit. Also, a UTM comes with automatic updates, which keep your system ready to combat the latest threats on the landscape.

Centralized integration and management

In a normal setup without UTM, you may have to juggle several security components at once, including a firewall, application control, a VPN, and others. This can take time and resources away from your team. However, with a UTM, you can consolidate everything and control it all with a single management console. This makes it easier

to monitor the system, as well as address particular components within the UTM that may need to be updated or checked.

The centralized nature of a UTM also allows you to monitor several threats simultaneously as they impact multiple components of your network. In a network without this centralized structure, when a multi-module attack is occurring, it can be very difficult to prevent it.

Cost-effectiveness

Because of its centralized setup, a UTM reduces the number of devices your organization needs to protect your network. This may result in significant cost savings. In addition, because fewer staff are required to monitor the system, you can save on manpower costs as well.

Increased awareness of network security threats

The combination of a UTM's centralization and faster operation results in an increased awareness of network security threats, enabling you to implement advanced threat protection (ATP). This equips your IT team to better manage advanced persistent threats (APTs) and other modern dangers on the landscape. The enhanced capability to address these kinds of threats comes from a UTM's ability to operate several threat response mechanisms in unison, which combine forces against the threats that attempt to infiltrate your network.

Faster security solution for businesses

With a UTM, you can streamline the way data is processed and use fewer resources at the same time. The UTM does not require as much resources as several components operating independent of each other. The higher efficiency you get from a UTM may allow you to free up resources to better manage other essential network-dependent processes.

Next-generation Firewalls vs. UTM (Unified Threat Management)

Although, on the surface, it may seem that the differences between next-generation firewalls (NGFWs) and UTM are merely semantic, depending on which NGFW you use, there may be some distinctions. To be clear, both solutions protect your network. With a UTM, however, there exists the possibility that you get services you do not need. Integrating these with your current network could involve extra work. It could also result in difficult decisions and a challenging setup process as you try to either combine the UTM's features with what you already have or pit one against the other to ascertain which solution is better.

With NGFWs, on the other hand, such as the Fortinet FortiGate, you can choose to turn on the features you need, making it a complete UTM solution. Conversely, you can choose to only use it as a firewall or activate some protections but not others. If, for example, you have FortiGate and choose to use it to its full capacity, it will also work as a UTM system.

Another difference is that an NGFW is an effective solution for larger enterprises, whereas a typical UTM may get overwhelmed by the demands of an enterprise.

threat management software